

John Kamau

AWS Cloud Security Engineer

Terraform · Zero Trust · IAM · CloudTrail · Auto Scaling

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PROFESSIONAL SUMMARY

AWS Cloud Security Engineer with hands-on experience designing and deploying production-grade, security-first infrastructure on AWS. Specializes in Zero Trust architecture, Infrastructure as Code with Terraform, and automated compliance using AWS Config. Built three progressive AWS security projects from the ground up — deployed entirely via CLI and Terraform on Ubuntu Linux — covering IAM, CloudTrail, VPC, EC2, Application Load Balancer, Auto Scaling, CloudWatch, SNS, and AWS Config. Background in Electrical and Electronics Engineering provides a strong foundation in systems thinking, problem analysis, and technical precision.

FEATURED PROJECTS



AWS Zero Trust Security Platform

May 2026

github.com/johntay379-hub/aws-zero-trust-security-platformzero-trust-alb-1441037960.us-east-1.elb.amazonaws.com

- Deployed 43 AWS resources with a single terraform apply command — IAM, S3, CloudTrail, VPC, ALB, ASG, CloudWatch, SNS, and AWS Config
- Built an Application Load Balancer across 2 availability zones with health checks every 30 seconds — zero downtime on instance failure
- Configured an Auto Scaling Group scaling from 1 to 4 EC2 instances based on CPU thresholds — fully automated, no human intervention
- Implemented AWS Config with 4 continuous compliance rules: S3 public access, EC2 IMDSv2, CloudTrail enabled, and root MFA
- Enforced IMDSv2 on all instances via Launch Template — blocks SSRF attacks on the metadata service
- EC2 instances accessible only through the ALB security group — no direct internet exposure



Terraform AWS Security Framework

May 2026

github.com/johntay379-hub/terraform-aws-security-framework

- Rebuilt the AWS CLI security framework as modular Terraform Infrastructure as Code — one command deployment and teardown
- Structured as 6 independent Terraform modules: IAM, S3, CloudTrail, VPC, EC2, CloudWatch — reusable across environments
- Enforced IMDSv2, least privilege IAM roles, encrypted S3 audit vault, and multi-region CloudTrail logging



AWS End-to-End Security Framework

April 2026

github.com/johntay379-hub/aws-end-to-end-security-framework100.49.157.164

- Deployed a complete 6-layer AWS security framework manually via AWS CLI on Ubuntu Linux — IAM, S3, CloudTrail, VPC, EC2, CloudWatch and SNS
- Configured multi-region CloudTrail with log file validation, encrypted S3 audit vault, and real-time SNS email alerting
- Ran 3 security simulation scenarios: unauthorized API access, SSH brute force, and log tampering — all blocked and logged

TECHNICAL SKILLS

Cloud

AWS — IAM · S3 · EC2 · VPC · CloudTrail · CloudWatch · SNS · ALB · ASG · AWS

	Config
IaC	Terraform — modular architecture · providers · state management · apply / destroy
Security	Zero Trust · Least Privilege · IMDSv2 · SSRF prevention · compliance monitoring
Networking	VPC · Subnets · Internet Gateway · Route Tables · Security Groups · NACLs
OS / CLI	Ubuntu Linux · AWS CLI · Bash scripting · Git · GitHub
Engineering	Electrical & Electronics Engineering · Systems thinking · Technical documentation

WORK EXPERIENCE

Electrical Engineer

Previous — Present

Previous Employer

- Designed, installed, and maintained electrical systems — developed a systematic, safety-first approach to technical problem solving
- Diagnosed and resolved complex electrical faults under pressure — directly transferable to troubleshooting cloud infrastructure issues
- Produced technical documentation and reports — foundation for the detailed project documentation in current cloud projects
- Collaborated with cross-functional teams on engineering projects — experience working in team environments

EDUCATION

Bachelor of Technology — Electrical and Electronics Engineering

Pursuing

RVINAP

PORTFOLIO

GitHub: github.com/johntay379-hub — 3 live AWS security projects with full documentation, architecture diagrams, and deployment proof